# Lab Report: Introduction to Machine Learning and Setting up the Environment

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**Lab No: 1**

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## 1. Objective

The objective of this lab is to:

- Understand the basic concept of Machine Learning.
- Learn about various ML types (Supervised, Unsupervised, Reinforcement Learning).
- Set up the ML programming environment using Python and essential libraries.

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## 2. Introduction

Machine Learning (ML) is a subfield of Artificial Intelligence (AI) that enables systems to learn from data and improve their performance without being explicitly programmed. ML algorithms are trained on data to make predictions or decisions. Real-world examples include spam filtering, facial recognition, recommendation systems, and disease diagnosis.

There are three main types of machine learning:

- **Supervised Learning:** Algorithms learn from labeled data.
- **Unsupervised Learning:** Algorithms find patterns in unlabeled data.
- **Reinforcement Learning:** Algorithms learn by interacting with an environment to achieve a goal.

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# 3. Required Tools and Environment

To perform ML tasks efficiently, we need to set up an appropriate programming environment. In this lab, we used the following tools:

## 3.1 Programming Language

- **Python** (due to its wide ML library support)

## 3.2 Libraries

- **NumPy:** For numerical operations
- **Pandas:** For data handling and analysis
- **Matplotlib/Seaborn:** For data visualization
- **Scikit-learn:** For implementing ML models

## 3.3 IDE/Platform

- **Jupyter Notebook** (An interactive environment for writing and executing code)

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# 4. Steps Performed

1. **Installed Python (version 3.x)** using Anaconda Distribution.
2. **Launched Jupyter Notebook** from the Anaconda Navigator.
3. **Installed required libraries** using the terminal:

```
bash
CopyEdit
pip install numpy pandas matplotlib seaborn scikit-learn
```

4. **Tested installation** by importing libraries in Jupyter:

```
python
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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
```

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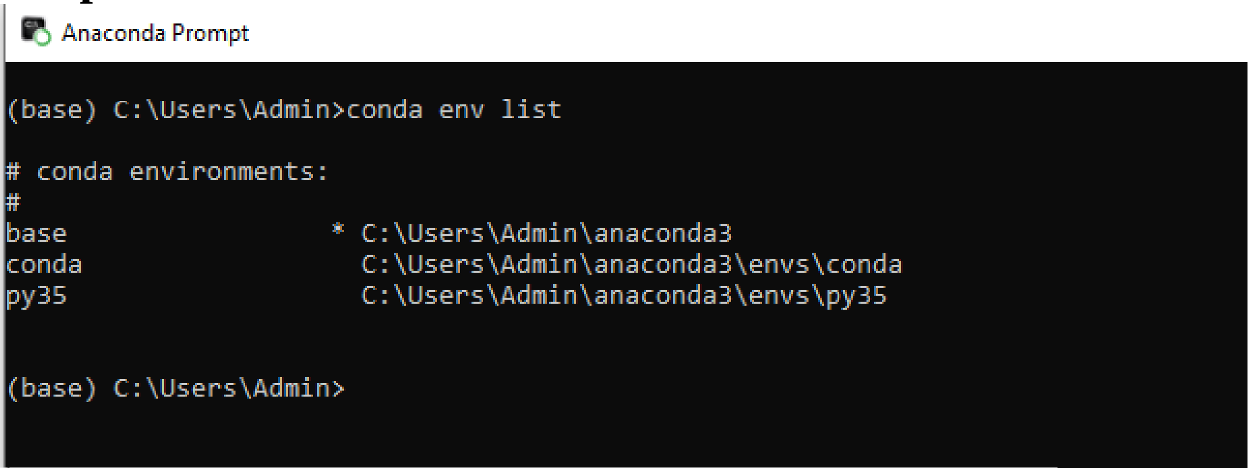
# 5. Output Screenshot

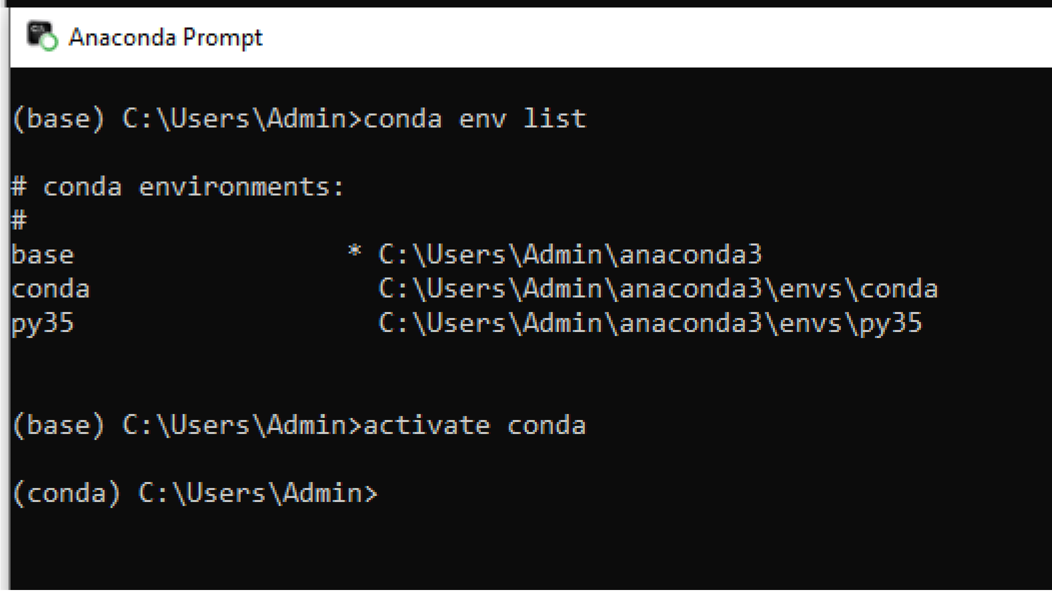


Anaconda Prompt

```
(base) C:\Users\Admin>conda env list

# conda environments:
#
base                  * C:\Users\Admin\anaconda3
conda                   C:\Users\Admin\anaconda3\envs\conda
py35                    C:\Users\Admin\anaconda3\envs\py35


(base) C:\Users\Admin>
```



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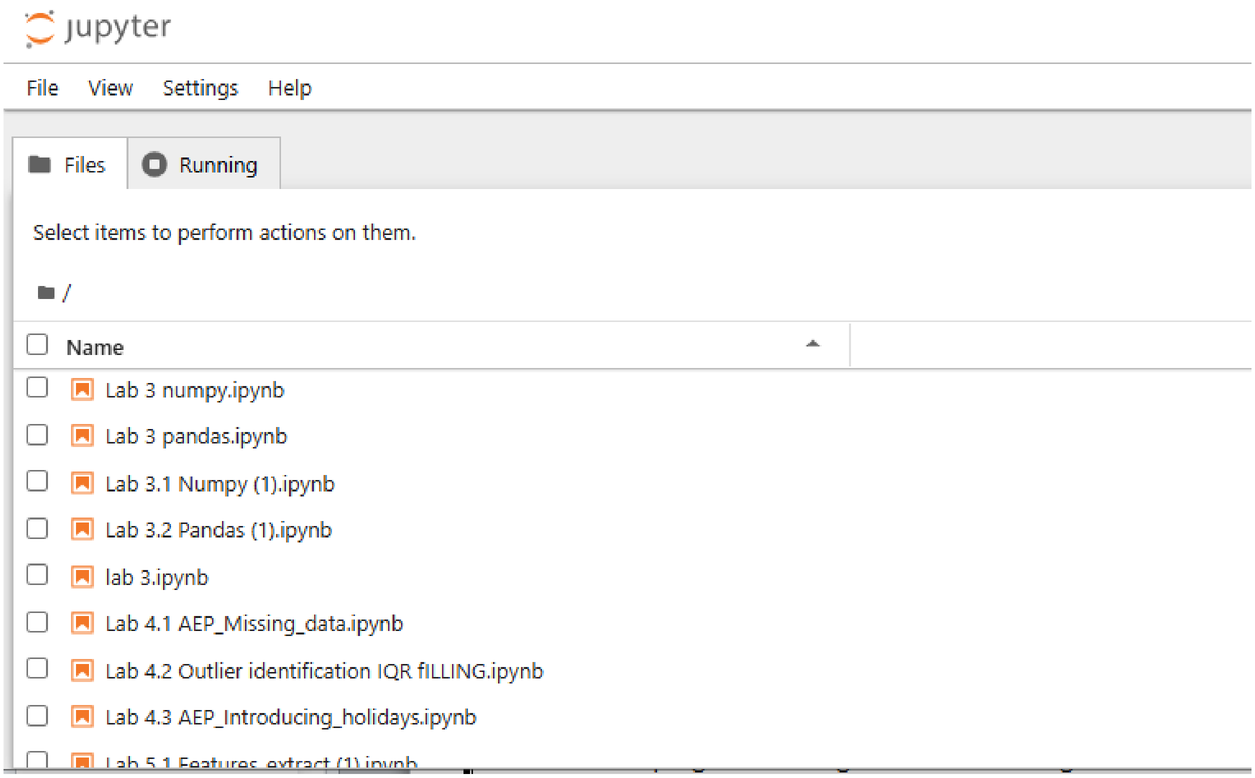
conda                   C:\Users\Admin\anaconda3\envs\conda
py35                    C:\Users\Admin\anaconda3\envs\py35


(base) C:\Users\Admin>activate conda

(conda) C:\Users\Admin>
```

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# 6. Conclusion

In this lab, we learned the basic idea of Machine Learning and successfully set up the ML environment using Python and popular libraries. This setup will be used for future labs to build, train, and test ML models.